

The differences between Hydroxocobalamin and Cyanocobalamin

Hydroxocobalamin and cyanocobalamin are two types of vitamins B12. Hydroxocobalamin is natural and used to treat cyanide poisoning is an anti-anemic class of drugs, while cyanocobalamin is synthetic and found in supplements and fortified foods.

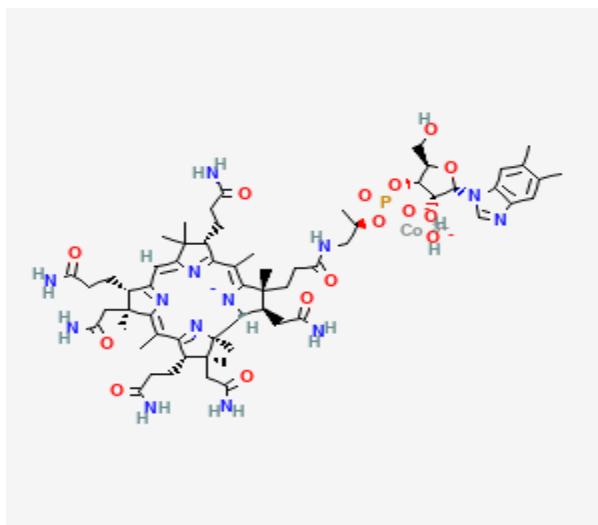
Generic name: Hydroxocobalamin

Brand name: Cyanokit

Chemical formula - $C_{62}H_{89}CON_{13}O_{15}P$

Molecular weight - 1346.4 g/mol

Structure -



Formulation – IM injection and IV solution for reconstitute (only for Cyanide poisoning treatment)

Indicated - treating vitamin B12 deficiency caused by dietary lack, malabsorption, insufficient intrinsic factor, intestinal parasites/bacteria, or poor B12 utilization. Treatment of cyanide poisoning (known or suspected).

Mechanism of Action - Hydroxocobalamin serves as a precursor to methyl-cobalamin and adenosyl-cobalamin, the active forms of vitamin B12. These cofactors are essential for DNA and amino acid synthesis, fatty acid metabolism, and the maintenance of nerve function. Furthermore, hydroxocobalamin plays a crucial role in the synthesis of purines and pyrimidines necessary for DNA replication and red blood cell formation.

Bioavailability and Absorption – High bioavailability with higher retention in body and it is readily absorbed in the distal half of the ileum.

Dose and administration –

Hydroxocobalamin - 1,000 mcg/mL solution IM

Cyanokit – 1000 mcg/ml IV using vented tubing infusion over 15 minutes or -Intraosseous may be used in pediatric patients and in adult patients when IV administration is not possible.

Storage - Solution for IM injection store at 20°C to 25°C (68°F to 77°F) and protect from light.

Side effects may cause – Diarrhea, Nausea, Vomiting, Abdominal pain, Injection site irritation, Heartburn, Flushing, Agitation, Acne

Severe side effects - Low potassium like muscle pain or weakness, muscle cramps, or an abnormal heartbeat, Kidney problems like unable to pass urine, blood in the urine, change in amount of urine passed, or weight gain, passing out, Black, tarry, or bloody stools, Trouble swallowing, Trouble with memory, Vision changes

Pricing –

Hydroxocobalamin Acetate IM solution 1000 mcg/mL (per mL): \$1.28

Cyanokit IV solution for reconstitute 5 g (per each): \$1,197.60

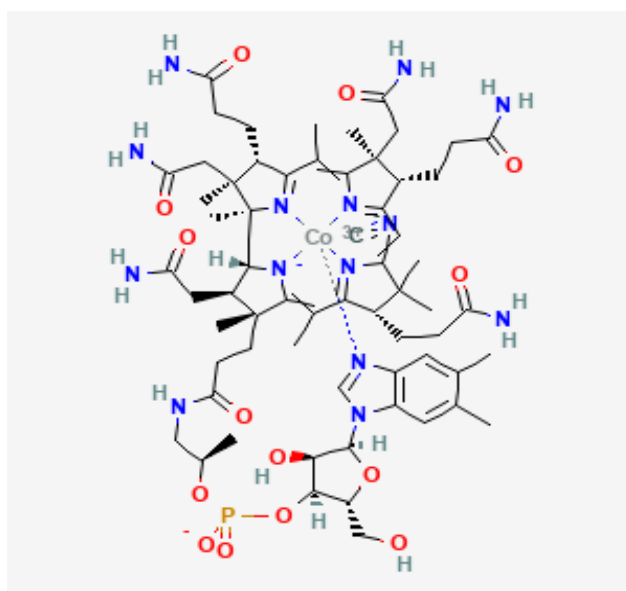
Generic name: **Cyanocobalamin**

Brand name: Dodex, Nascobal, Vitamin Deficiency System-B12, B-12 Compliance Injection

Chemical formula – $C_{63}H_{88}CoN_{14}O_{14}P$

Molecular weight – 1355.4 g/mol

Structure –



Formulation – IM, deep subq, Oral tablets, Sublingual tablets, Intranasal

Indication – Treatment of pernicious anemia or vitamin B₁₂ deficiency due to dietary deficiencies or to prevention of vitamin B₁₂ deficiency in patients with increased B₁₂ requirements (e.g.- due to pregnancy, thyrotoxicosis, hemorrhage, malignancy, liver or kidney disease).

Mechanism of Action – The absorption of cyanocobalamin occurs through the small intestine after it binds to intrinsic factors and other cobalamin-binding proteins. When administered parenterally, the vitamin enters the bloodstream immediately. In the blood, vitamin B12 attaches to plasma proteins. Tissues absorb vitamin B12 via specific binding proteins, transcobalamin I and II, which facilitate its

transport into cells. The majority of the vitamins are stored in the liver. Vitamin B12 is crucial for DNA synthesis and energy production, particularly in erythroid progenitor cells.

Bioavailability and Absorption – cyanocobalamin is rapidly absorbed from injection sites and reaches its peak within 1 hour after intramuscular (IM) injection and primarily excreted through the kidney. Approximately 50% to 98% of the injected cyanocobalamin is in the urine. A significant portion is excreted within the first 8 hours. Oral formulation absorbed in the small intestine and have very low bioavailability because of the presence of low intrinsic factor, which is necessary to bind to vitamin B12 and facilitate its absorption into the bloodstream.

Dose and administration – The cyanocobalamin content of over-the-counter products varies between 50-1000 mcg oral tablets can be administer without regard of food, IM, SubQ 100-1000 mcg once daily for up to 7 days is recommended according to severity of Vit B 12 deficiency. Nasal spray (Nascobal) must be administered 1 hour before or 1 hour after the ingestion of hot foods/liquids.

Storage – Injection store between 20°C and 25°C (68°F and 77°F) and protect from light, and Intranasal spray store at 15°C to 30°C (59°F to 86°F); do not freeze. Protect from light

Side effects – Headache, Asthenia, Paresthesia, Glossitis, Nausea, Pruritus, Hypokalemia, Diarrhea, Pulmonary edema, Swelling

Pricing –

Cyanocobalamin Sublingual liquid 5000 mcg/mL (per mL): \$0.11

Cyanocobalamin Injection 1000 mcg/mL (per mL): \$1.21 - \$8.74

Cyanocobalamin Nasal 500 mcg/0.1 mL (per each): \$191.51 and Nascobal Nasal 500 mcg/0.1 mL (per each): \$212.79

Vitamin B-12 ER Oral tablets 1000 mcg (per each): \$0.07 and Vitamin B-12 Oral tablets 1000 mcg (per each): \$0.04 - \$0.22

Reference:

[Ramezanpour Ahangar E, Annamaraju P. Hydroxocobalamin. [Updated 2023 May 29]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.]

<https://www.ncbi.nlm.nih.gov/books/NBK557632/>

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[Vasavada A, Patel P, Sanghavi DK. Cyanocobalamin. [Updated 2024 Jan 31]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-]

<https://www.ncbi.nlm.nih.gov/books/NBK555964/#article-88826.s1>

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